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OOP ABAP Reports

Deloitte Technology Academy (DTA)

Agenda

Topic	Descriptions	Duration
Object-Oriented Programming (OOP) Reports	Implementing Reports Using Advanced Business Application Programming (ABAP) List Viewer (ALV) Function Module (FM), and ALV Using Class CL_GUI_ALV_GRID	4 hours
ALV Object Model (SALV)	ALV Using ALV Object model (SALV)	

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Learning Objectives

By the end of this session, you will be able to:

- Describe the advantages of creating ALV reports using CL_GUI_ALV_GRID class
- Explain how to create a simple ALV report using CL_GUI_ALV_GRID class
- Describe the new object model—ECC 6.0
- Explain programming with class/methods and list out the main classes in SALV—GRID
- Generate ALV reports using OOP

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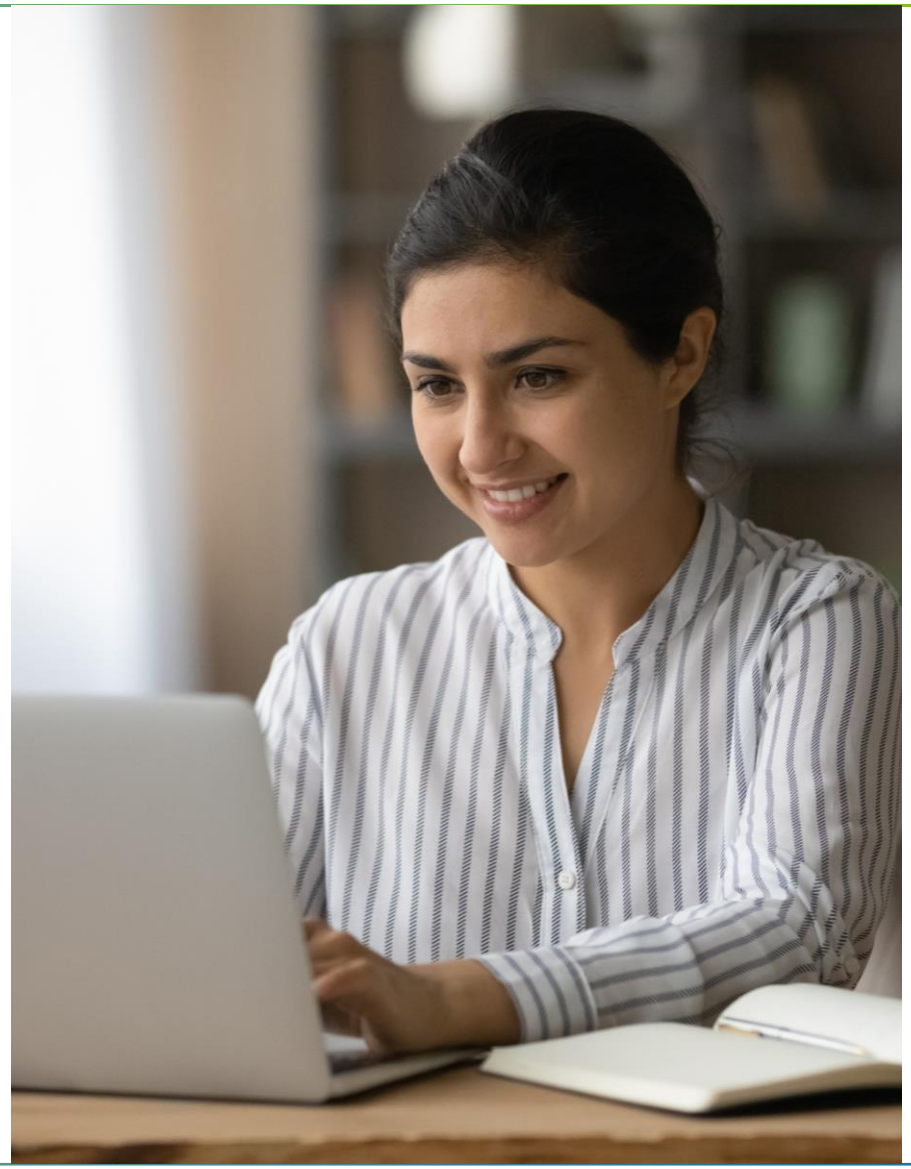
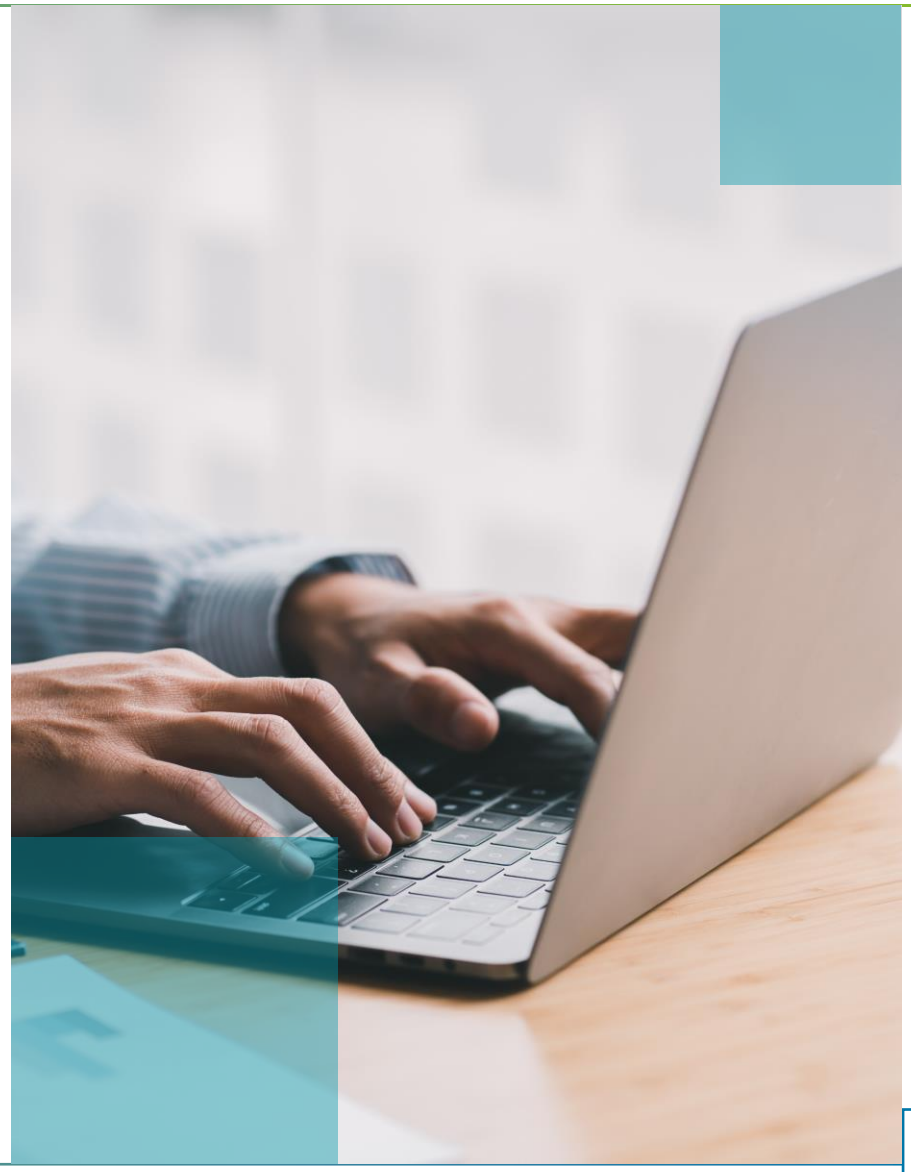
Object-Oriented Programming (OOP) Reports



Prerequisites

- Basic knowledge/concepts of object-oriented programming
  - Usage of class methods
- Basic Dynpro (screen) Programming
  - Creating and calling a screen
- ABAP programming
  - Creating classical and ALV reports



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Advantages of Creating ALV Using CL_GUI_ALV_GRID Class

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Methods

Basic Methods	CONSTRUCTOR, SET_TABLE_FOR_FIRST_DISPLAY
Elements of Grid Control	GET_CURRENT_CELL, GET_SELECTED_ROWS
Layout Handling	GET_FRONTEND_FIELDCATALOG, SET_FRONTEND_LAYOUT
Generic Functions	GET_SUBTOTALS, GET_VARIANT
Event Handling	SET_TOOLBAR_INTERACTIVE
Interfaces	OFFLINE, ACTIVATE_REPREP_INTERFACE

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Events

User Defined Text Output

PRINT_END_OF_LIST,
PRINT_TOP_OF_LIST

Self Defined and Standard Functions

USER_COMMAND,
BEFORE_USER_COMMAND

Mouse Controlled Actions in Grid Control

BUTTON_CLICK,
DOUBLE_CLICK,
HOTSPOT_CLICK

Definition of Self Defined Functions

TOOLBAR, MENU_BUTTON

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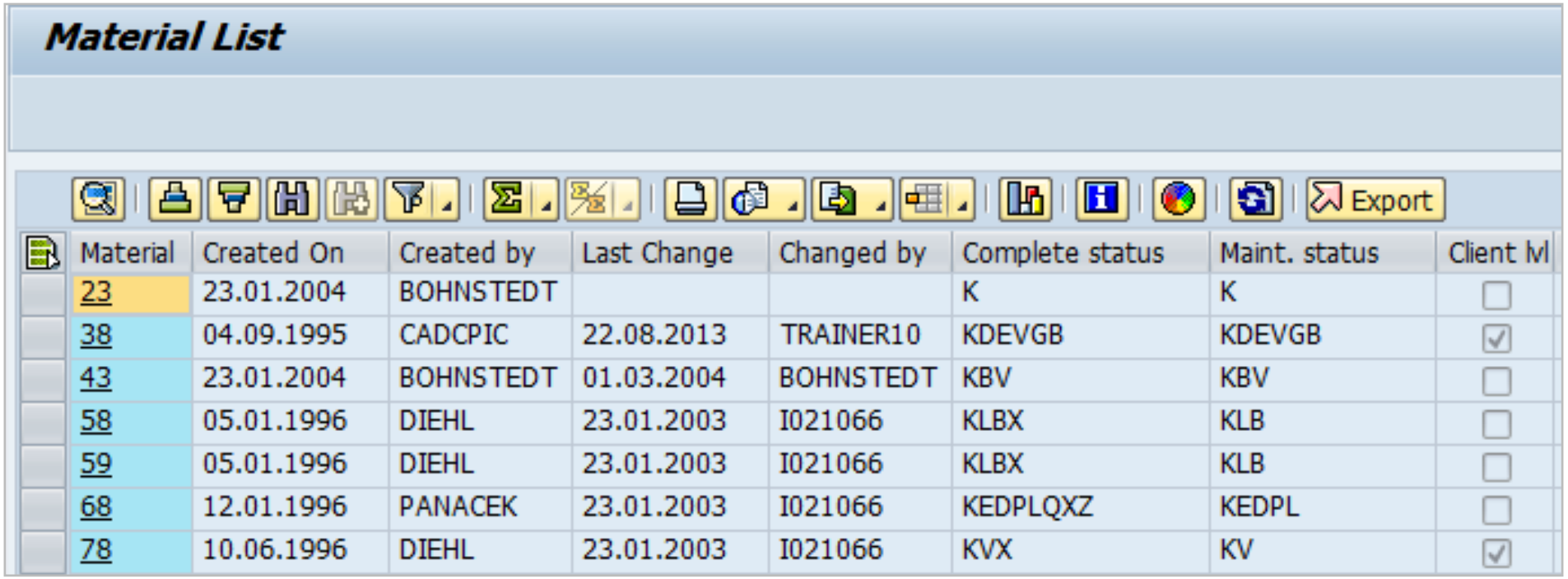
Creating a Simple ALV Report Using CL_GUI_ALV_GRID Class

- Input and data fetch
- Local class definition and implementation of custom methods using events
- Screen and container addition
- Display ALV



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43	23.01.2004	BOHNSTEDT	01.03.2004	BOHNSTEDT	KBV	KBV	☐
58	05.01.1996	DIEHL	23.01.2003	I021066	KLBX	KLB	☐
59	05.01.1996	DIEHL	23.01.2003	I021066	KLBX	KLB	☐
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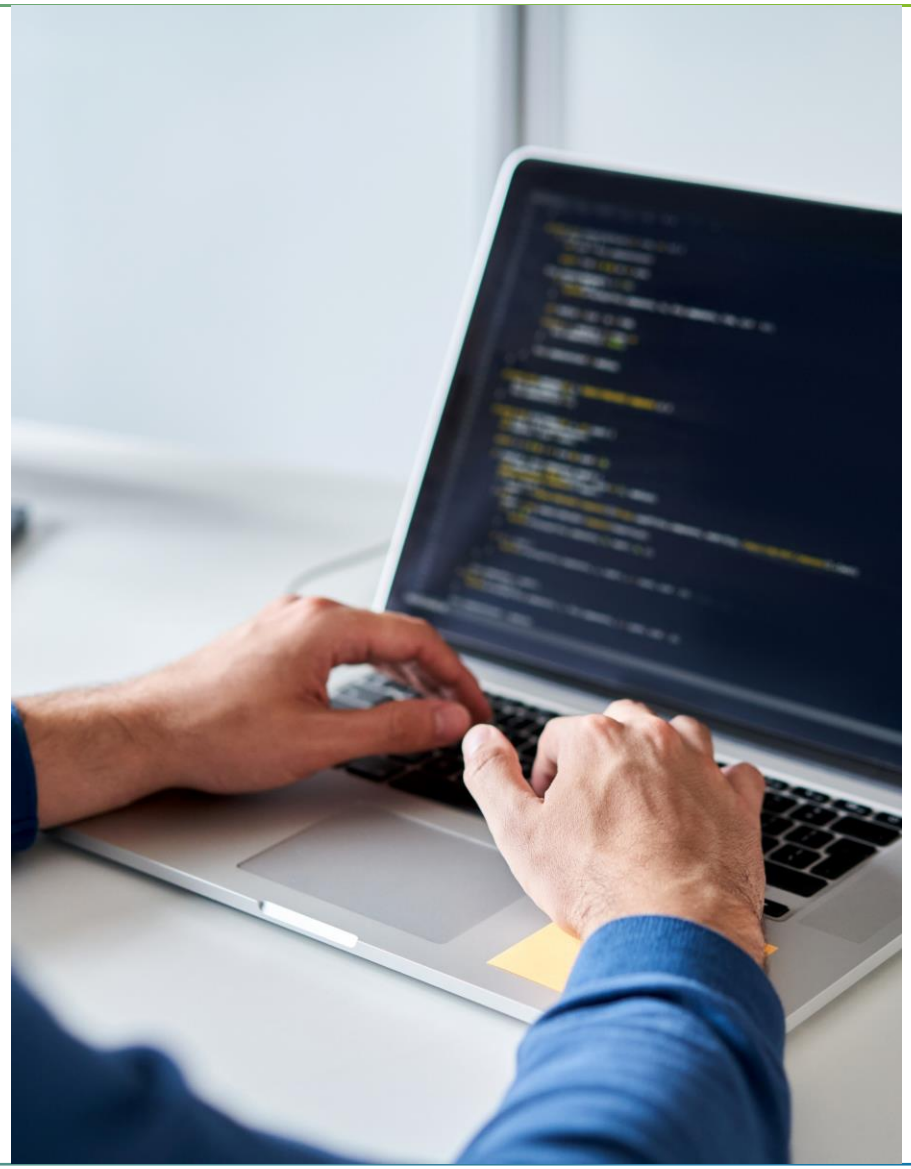
ALV Object Model (SALV)



Usage of ALV Function Modules/Class

One function module/class for each flavor

Callbacks for applications individually coded top/end of list/page

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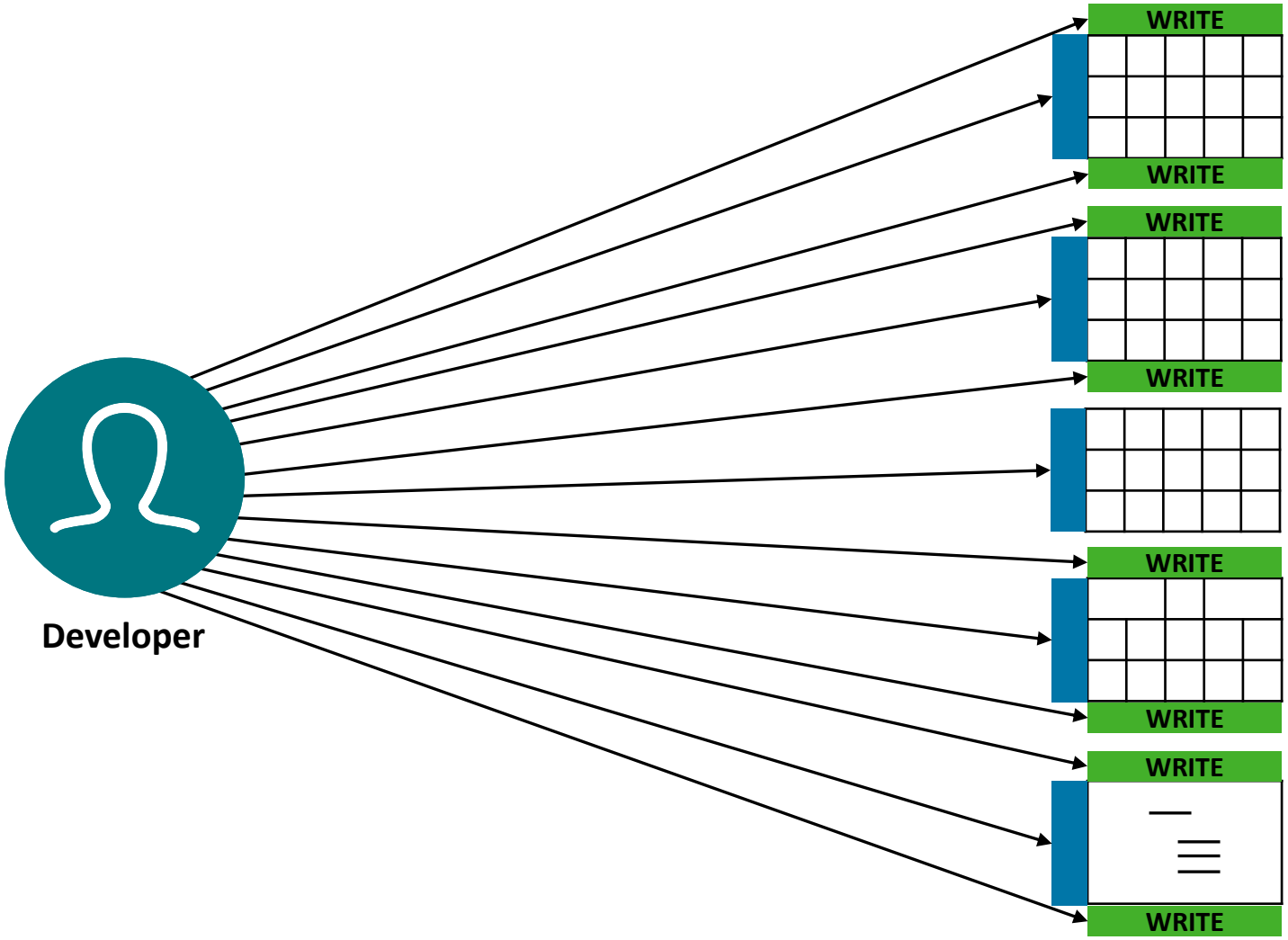
Reporting Using ALV GRID in 4.6C



ALV List

REUSE_ALV_LIST_DISPLAY

Grid Fullscreen

REUSE_ALV_GRID_DISPLAY

ALV Grid

CL_GUI_ALV_GRID

Hier.-Sequ.List

RESUE_ALV_HIERSEQULIST_DISPLAY

ALV Tree

CL_GUI_ALV_TREE

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SAP NetWeaver 2004

Optimized, consistent, robust API for the presentation of data for all ALV flavors

Completely data and meta data driven interface for top/end of list/page areas

Single class for handling tabular data

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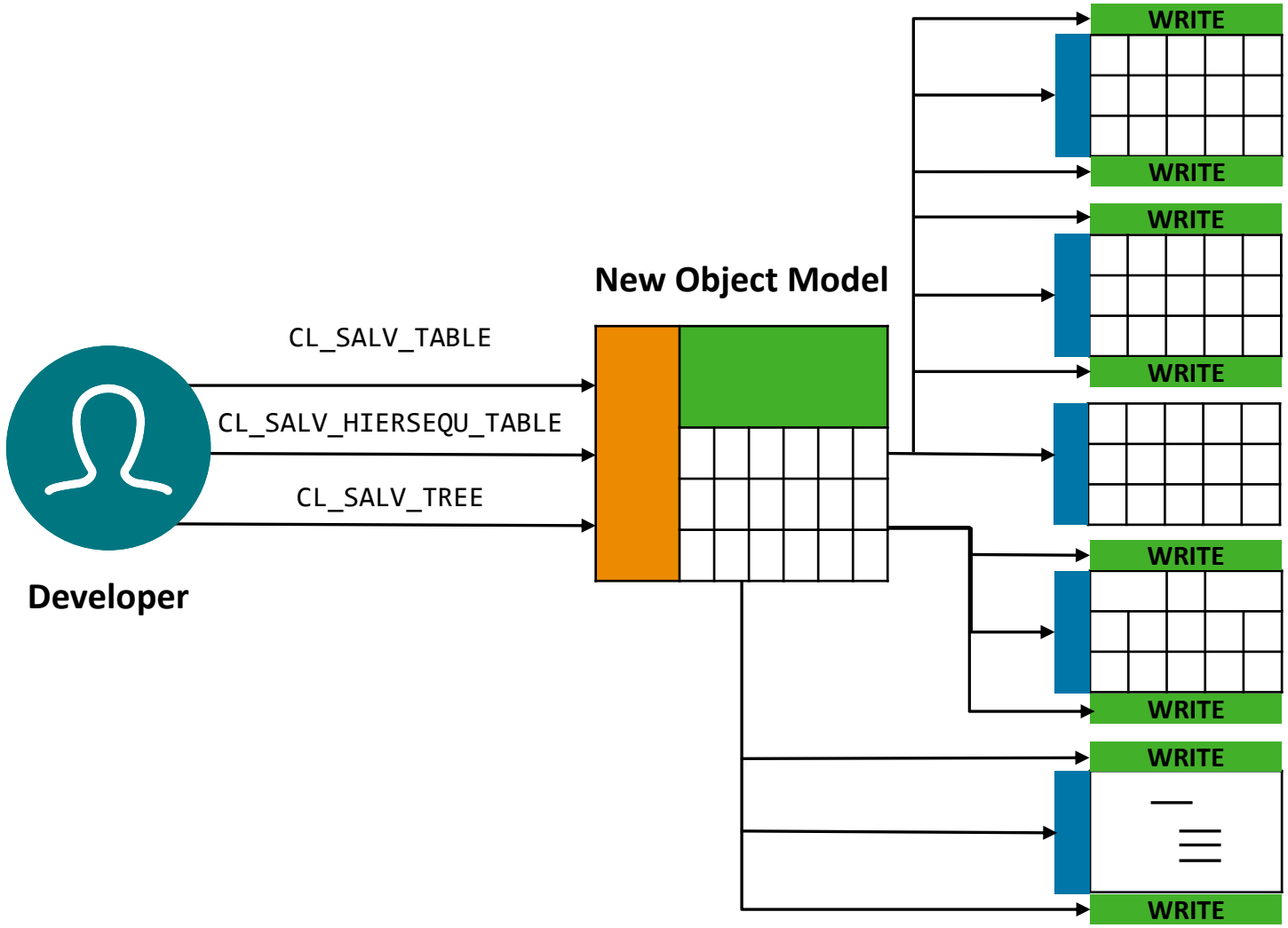
The New Object Model—ECC 6.0



ALV List

REUSE_ALV_LIST_DISPLAY

Grid Fullscreen

REUSE_ALV_GRID_DISPLAY

ALV Grid

CL_GUI_ALV_GRID

Hier.-Sequ.List

RESUE_ALV_HIERSEQULIST_DISPLAY

ALV Tree

CL_GUI_ALV_TREE

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Classes—ABAP Objects

ABAP objects provide an advanced level of encapsulation that improves the maintainability and stability of programs

ABAP objects allow you to instantiate multiple instances of a single class

ABAP objects greatly enhance code reuse through “inheritance” a key feature of Object Orientation (OO)

ABAP objects are the only way to use new ABAP technology—Web Dynpro ABAP, etc.

Explicit event handling

Better interfaces

ABAP Objects

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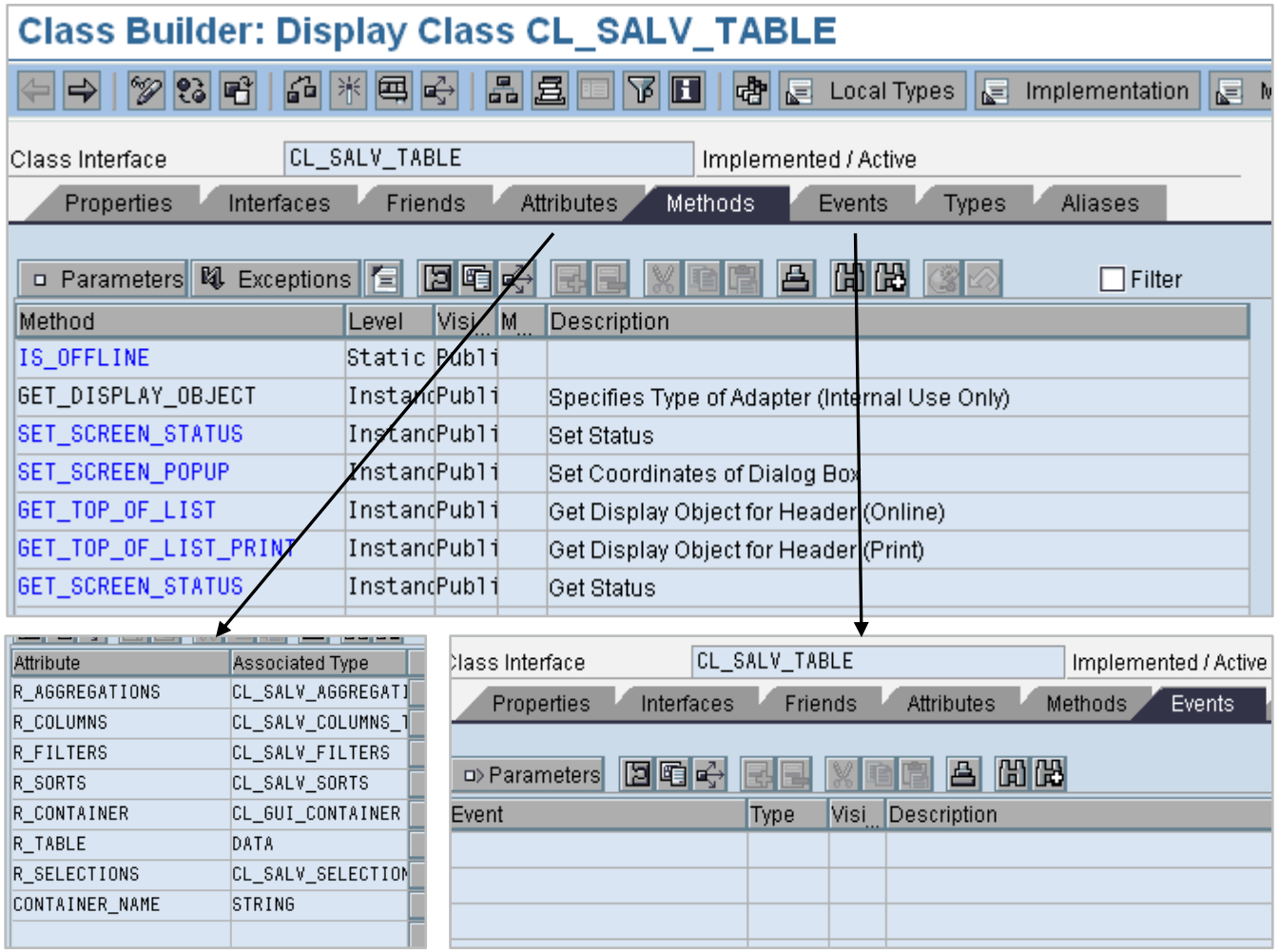
ABAP Class—Transaction SE24



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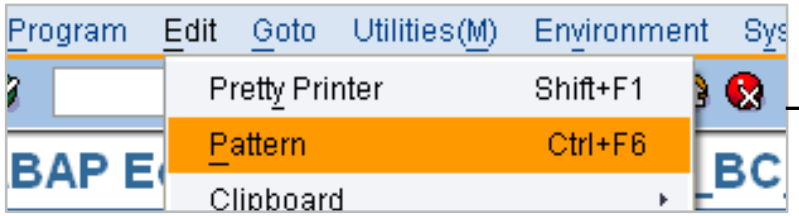
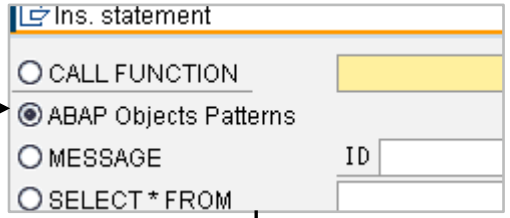
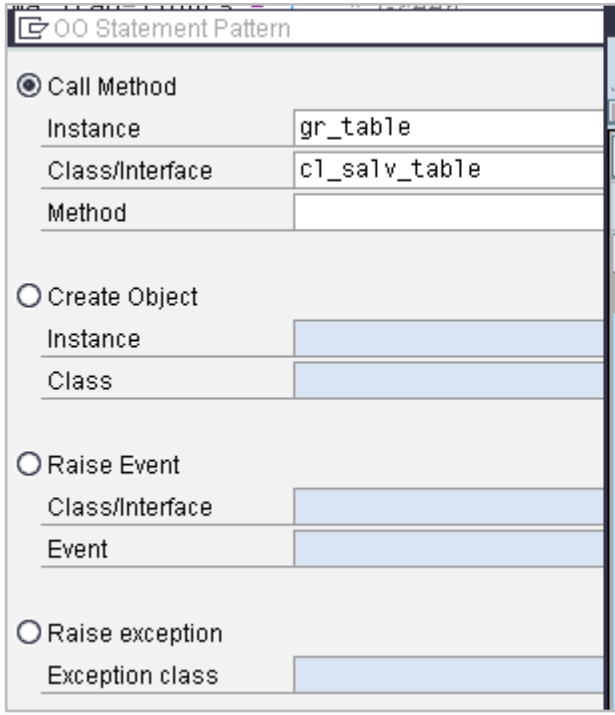
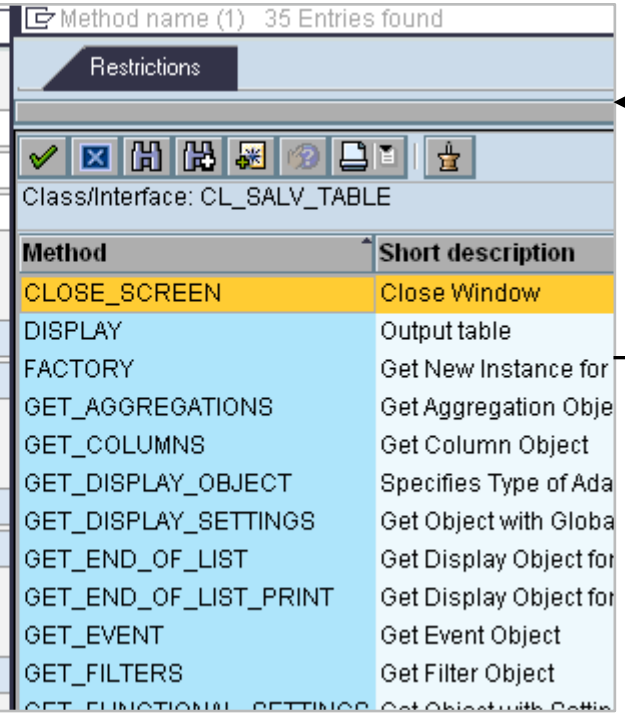
Programming With Class/Methods

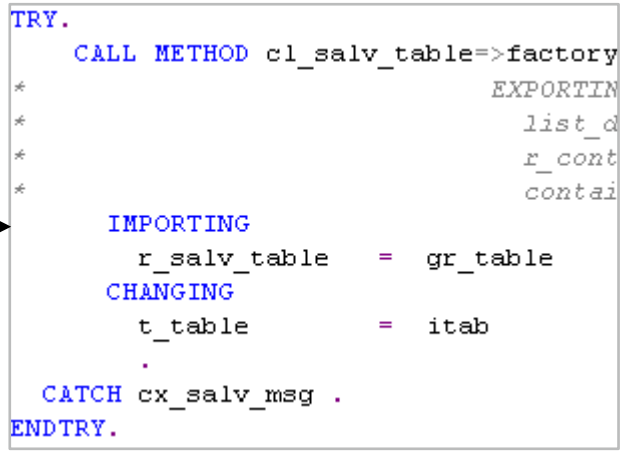
In SE38–ABAP Editor











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CL_SALV_TABLE

Description
Two-dimensional data display in ALV_GRID form
Object orientation, or to be more precise, object-oriented programming
No screens to define, no field catalogs to create, no Process Before Output (PBO), and Process After Output (PAI) modules, the standard menu bar can be used
More functionality for application development
Extremely fast delivery to users
A way to get used to what ABAP programming is trending towards
Only available in ECC 6.0 and above

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Main Classes in SALV—GRID

Main Class	CL_SALV_TABLE
Functions	CL_SALV_FUNCTIONS
Display Settings	CL_SALV_DISPLAY_SETTINGS
Columns	CL_SALV_COLUMNS_TABLE and CL_SALV_COLUMN_TABLE
Sorts	CL_SALV_SORTS

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Main Classes in SALV—GRID

Aggregations	CL_SALV_AGGREGATIONS
Filters	CL_SALV_FILTERS
Layouts	CL_SALV_LAYOUT
Events	CL_SALV_EVENTS_TABLE

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CL_SALV_TABLE—Main Class for ALV_GRID

</>

DATA:

```
GR_TABLE TYPE REF TO CL_SALV_TABLE.
CL_SALV_TABLE=>FACTORY( IMPORTING R_SALV_TABLE = GR_TABLE CHANGING T_TABLE = ITAB ).
GR_TABLE->DISPLAY( ).
```


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CL_SALV_FUNCTIONS—Adds Functions to Toolbar

```
DATA:
GR_FUNCTIONS TYPE REF TO CL_SALV_FUNCTIONS.
GR_FUNCTIONS = GR_TABLE->GET_FUNCTIONS( ).
GR_FUNCTIONS->SET_ALL( ABAP_TRUE ).
```

Cli...	Air...	Flig...	Co...	Depart. city	De...	Co...	Arrival city	Ta...	Flight t...	Departure
0	AA	17	US	NEW YORK	JFK	US	SAN FRANCISCO	SFO	6:01	11:00:00
000	AA	64	US	SAN FRANCISCO	SFO	US	NEW YORK	JFK	5:21	09:00:00
000	AZ	555	IT	ROME	FCO	DE	FRANKFURT	FRA	2:05	19:00:00
000	AZ	788	IT	ROME	FCO	JP	TOKYO	TYO	12:55	12:00:00
000	AZ	789	JP	TOKYO	TYO	IT	ROME	FCO	15:40	11:45:00
000	AZ	790	IT	ROME	FCO	JP	OSAKA	KIX	13:35	10:35:00
000	DL	106	US	NEW YORK	JFK	DE	FRANKFURT	FRA	7:55	19:35:00
000	DL	1699	US	NEW YORK	JFK	US	SAN FRANCISCO	SFO	6:22	17:15:00
000	DL	1984	US	SAN FRANCISCO	SFO	US	NEW YORK	JFK	5:25	10:00:00

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CL_SALV_DISPLAY_SETTINGS

```
DATA:
GR_DISPLAY TYPE REF TO CL_SALV_DISPLAY_SETTINGS.
GR_DISPLAY = GR_TABLE->GET_DISPLAY_SETTINGS( ).
GR_DISPLAY->SET_STRIPED_PATTERN( CL_SALV_DISPLAY_SETTINGS=>TRUE ).
GR_DISPLAY->SET_LIST_HEADER( 'THIS IS THE HEADING' ) .
```

Cli...	Air...	Flig...	Co...	Depart. city	De...	Co...	Arrival city	Ta...	Flght t...	Departure
0	AA	17	US	NEW YORK	JFK	US	SAN FRANCISCO	SFO	6:01	11:00:00
000	AA	64	US	SAN FRANCISCO	SFO	US	NEW YORK	JFK	5:21	09:00:00
000	AZ	555	IT	ROME	FCO	DE	FRANKFURT	FRA	2:05	19:00:00
000	AZ	788	IT	ROME	FCO	JP	TOKYO	TYO	12:55	12:00:00
000	AZ	789	JP	TOKYO	TYO	IT	ROME	FCO	15:40	11:45:00
000	AZ	790	IT	ROME	FCO	JP	OSAKA	KIX	13:35	10:35:00
000	DL	106	US	NEW YORK	JFK	DE	FRANKFURT	FRA	7:55	19:35:00
000	DL	1699	US	NEW YORK	JFK	US	SAN FRANCISCO	SFO	6:22	17:15:00
000	DL	1984	US	SAN FRANCISCO	SFO	US	NEW YORK	JFK	5:25	10:00:00

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Make Column Settings—Old Field Catalog

</>

Columns—CL_SALV_COLUMNS_TABLE and CL_SALV_COLUMN_TABLE

```
DATA: GR_COLUMNS TYPE REF TO CL_SALV_COLUMNS_TABLE.
DATA: GR_COLUMN TYPE REF TO CL_SALV_COLUMN_TABLE.
GR_COLUMNS = GR_TABLE->GET_COLUMNS( ).
GR_COLUMN ?= GR_COLUMNS->GET_COLUMN( 'CITYTO' ).
GR_COLUMN->SET_LONG_TEXT( 'THIS IS LONG TEXT' ).
GR_COLUMN->SET_MEDIUM_TEXT( 'THIS IS MED TEXT' ).
GR_COLUMN->SET_SHORT_TEXT( 'THIS IS SH' ).
GR_COLUMN ?= GR_COLUMNS->GET_COLUMN( 'CITYFROM' ).
COLOR-COL = '6'. COLOR-INT = '1'. COLOR-INV = '0'.
GR_COLUMN->SET_COLOR( COLOR ).
```


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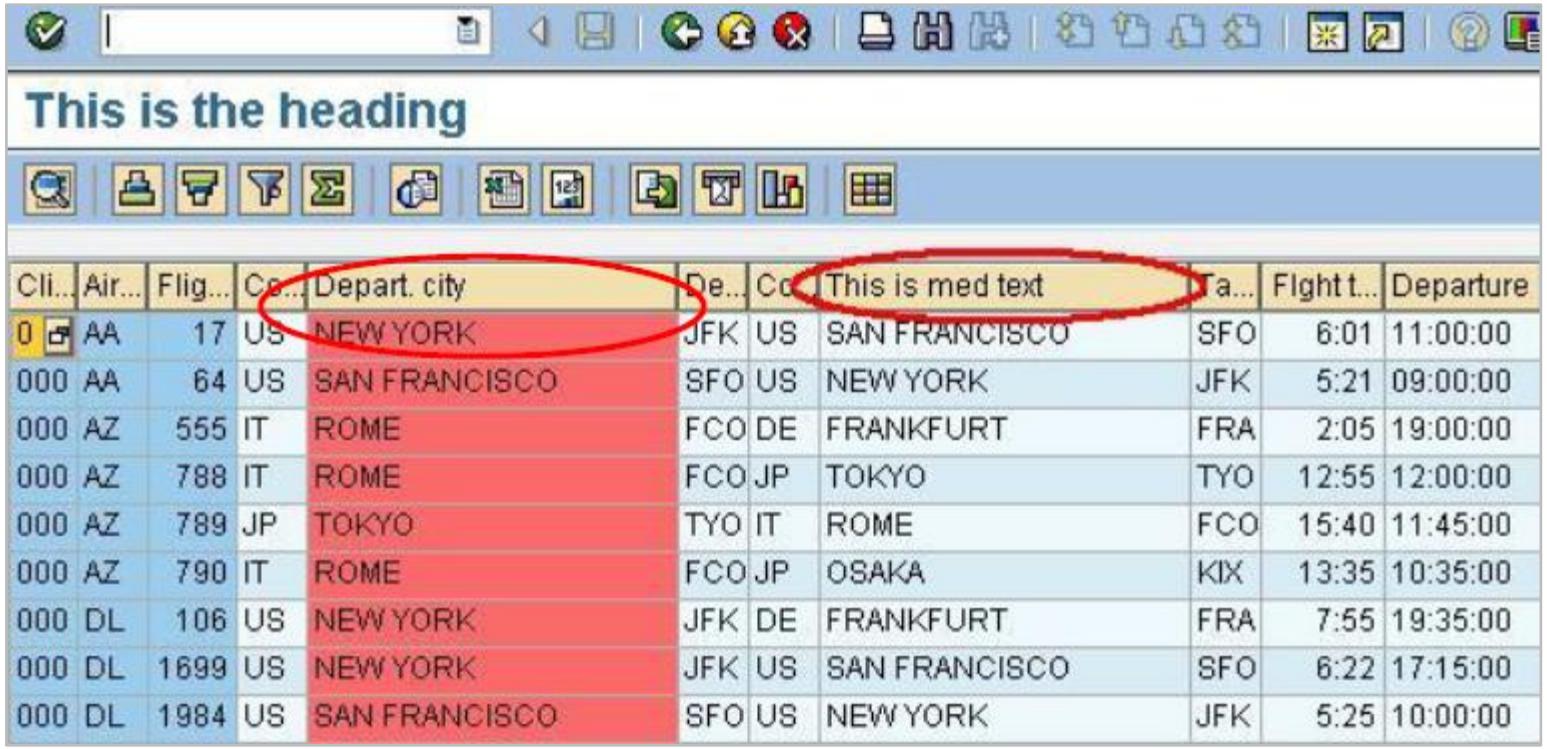
Columns—CL_SALV_COLUMNS_TABLE and CL_SALV_COLUMN_TABLE



This is the heading

Cli...	Air...	Flig...	Co...	Depart. city	De...	Co	This is med text	Ta...	Flght t...	Departure
0	AA	17	US	NEW YORK	JFK	US	SAN FRANCISCO	SFO	6:01	11:00:00
000	AA	64	US	SAN FRANCISCO	SFO	US	NEW YORK	JFK	5:21	09:00:00
000	AZ	555	IT	ROME	FCO	DE	FRANKFURT	FRA	2:05	19:00:00
000	AZ	788	IT	ROME	FCO	JP	TOKYO	TYO	12:55	12:00:00
000	AZ	789	JP	TOKYO	TYO	IT	ROME	FCO	15:40	11:45:00
000	AZ	790	IT	ROME	FCO	JP	OSAKA	KIX	13:35	10:35:00
000	DL	106	US	NEW YORK	JFK	DE	FRANKFURT	FRA	7:55	19:35:00
000	DL	1699	US	NEW YORK	JFK	US	SAN FRANCISCO	SFO	6:22	17:15:00
000	DL	1984	US	SAN FRANCISCO	SFO	US	NEW YORK	JFK	5:25	10:00:00

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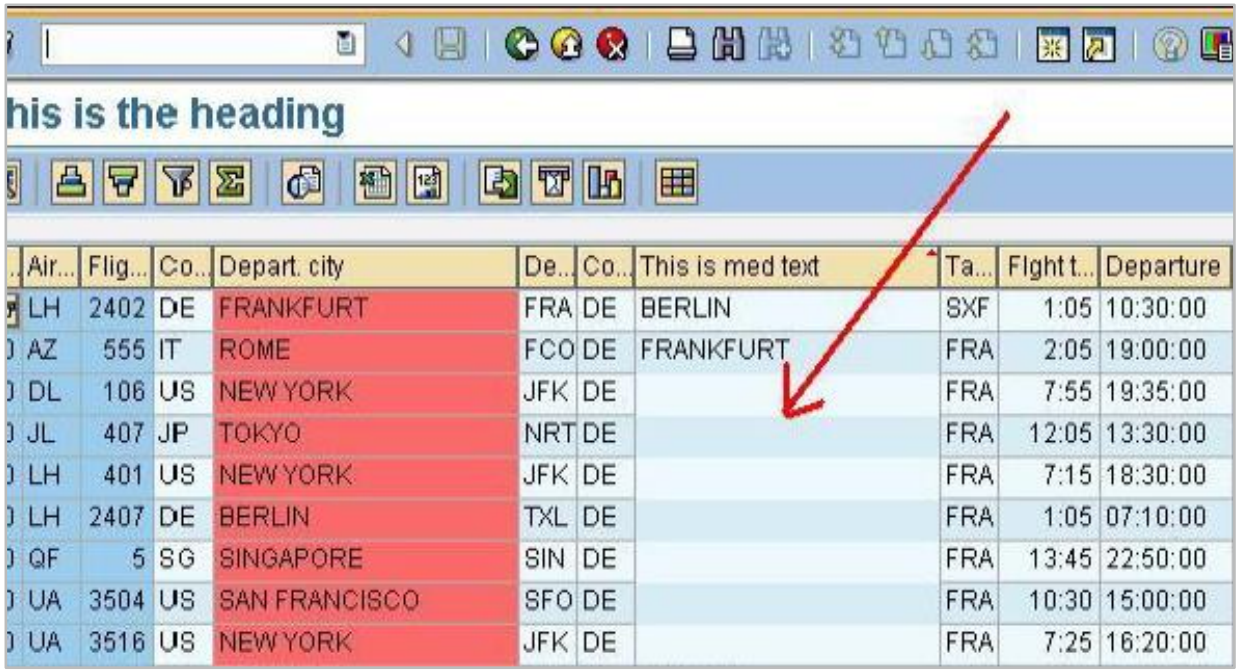
CL_SALV_SORTS—Sort Data by Defined Column

```
DATA: GR_SORTS TYPE REF TO CL_SALV_SORTS.
GR_SORTS = GR_TABLE->GET_SORTS( ).
GR_SORTS->ADD_SORT( COLUMNNAME = 'CITYTO' ).
```



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CL_SALV_AGGREGATIONS—“Do Sum” for Field

```
DATA: GR_AGG TYPE REF TO CL_SALV_AGGREGATIONS.
GR_SORTS->ADD_SORT( COLUMNNAME = 'CITYTO'
SUBTOTAL = ABAP_TRUE).
GR_AGG = GR_TABLE->GET_AGGREGATIONS( ).
GR_AGG->ADD_AGGREGATION( 'DISTANCE' ).
```

Air...	Flig...	Co...	Depart. city	De...	Co...	This is med text	Ta...	Distance	Dis
LH	2402	DE	FRANKFURT	FRA	DE	BERLIN	SXF	555	KM
						BERLIN		**555**	**KM**
AZ	555	IT	ROME	FCO	DE	FRANKFURT	FRA	845	MI
DL	106	US	NEW YORK	JFK	DE		FRA	3,851	MI
JL	407	JP	TOKYO	NRT	DE		FRA	9,100	KM
LH	401	US	NEW YORK	JFK	DE		FRA	6,162	KM
LH	2407	DE	BERLIN	TXL	DE		FRA	555	KM
QF	5	SG	SINGAPORE	SIN	DE		FRA	10,000	KM
UA	3504	US	SAN FRANCISCO	SFO	DE		FRA	5,685	MI
UA	3516	US	NEW YORK	JFK	DE		FRA	6,162	KM
						FRANKFURT		**31,979**	**KM**

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CL_SALV_LAYOUT—Manage/Save Layout

```
DATA: GR_LAYOUT TYPE REF TO CL_SALV_LAYOUT.
GR_LAYOUT = GR_TABLE->GET_LAYOUT( ).
KEY-REPORT = SY-REPID.
GR_LAYOUT->SET_KEY( KEY ).
GR_LAYOUT->SET_SAVE_RESTRICTION( CL_SALV_LAYOUT=>RESTRICT_NONE ).
```

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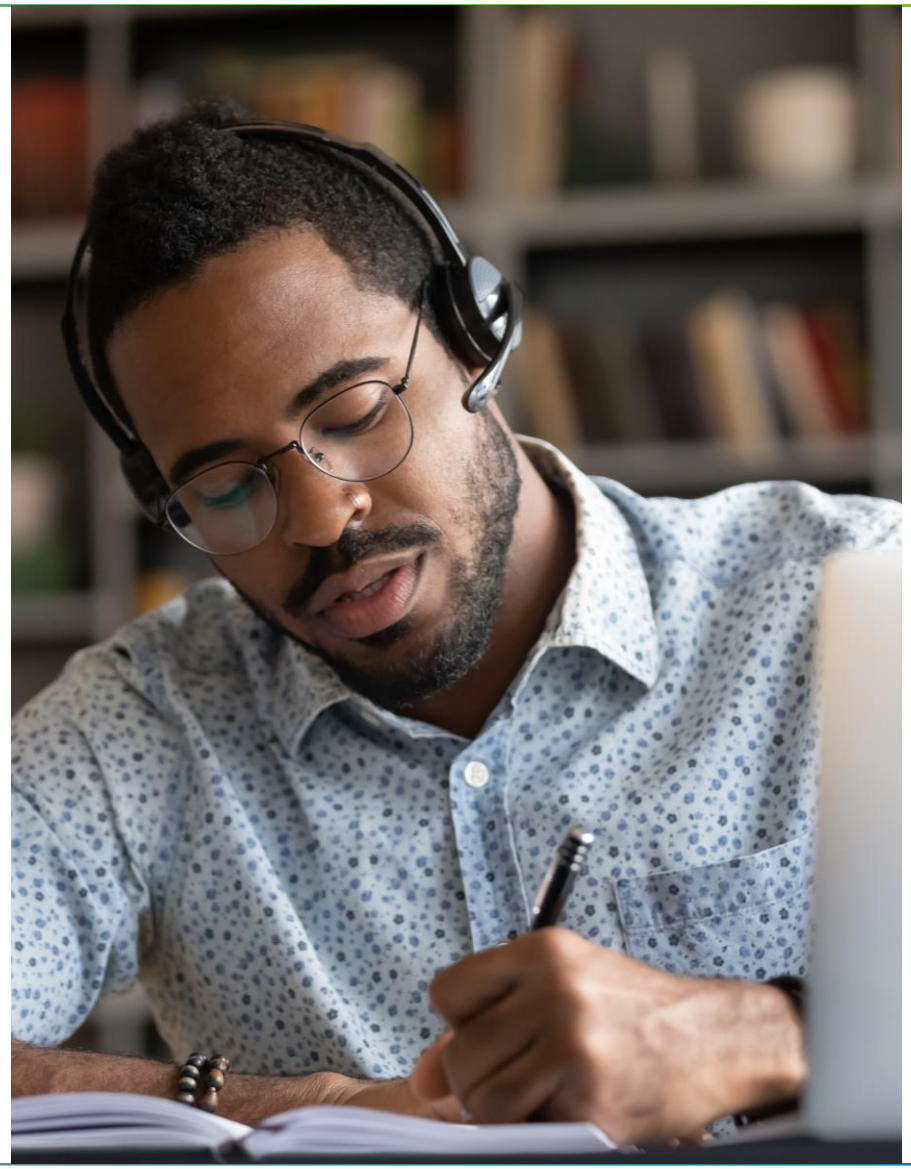
Hands-On Labs



Hands-On Activity 1

</>

Activity Details: Create an executable program with a selection screen

Problem Statement

Parameters

- P_ AUART type VBAK- AUART

Select Options

- S_SALES type VBAK-VBELN
- S_ERDAT type VBAK-ERDAT

- S_ERDAT needs to have a default value of 14 days prior to the current date and current date.
- Validate the selection screen parameters, P_AUART, and select options and S_SALES.
- Display appropriate error messages in case of an invalid entry, based on the selected radio button.
- On execution of the report, fetch all the records from tables VBAK that match the corresponding selection criteria.

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Hands-On Activity 1 (Cont.)

Activity Details: Create an executable program with a selection screen	
Problem Statement	• Create an executable program and extract the data from the database table based on the selection screen parameters. • Create a screen 100 and mark an area for the custom container control in the graphical Screen Painter (icon identified by letter 'C'). Assign the name CCCONTAINER to this area. • Declare reference variables for the ALV Grid Control (CL_GUI_ALV_GRID) and the container (CL_GUI_CUSTOM_CONTAINER). • In the PBO module of the screen, you must now instantiate the container control and the ALV Grid Control. By doing this, you create a link between the container control and the screen, using the container created in the Screen Painter. • Use the FM ‘LVC_FIELDCATALOG_MERGE’ for generating the field catalog by passing the structure name.

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Hands-On Activity 1 (Cont.)

Activity Details: Create an executable program with a selection screen	
Problem Statement	• **Event Handling**: Create a local class with the name ‘LCL_ALV_EVENTS’. • Define a method ‘HANDLE_HOTSTOP_CLICK’ to handle the event ‘HOTSPOT_CLICK’ of the class ‘CL_GUI_ALV_GRID’. The method should have an importing parameter ‘E_ROW_ID’. • Implement the method ‘HANDLE_HOTSPOT_CLICK’. • Read the sales order number based on the row index. • Set the Parameter ID ‘AUN’ with the sales order number. • Call the transaction VA03 by skipping the initial screen. • Register the event by using the statement SET HANDLER.

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Hands-On Activity 1 (Cont.)

Activity Details: Create an executable program with a selection screen	
Output	Display the report in the below format:

Sales order	Document type	Sales org.	Dist. Channel	Division	Creation date	Creator
VBAK-VBELN	VBAK-AUART	VBAK-VKORG	VBAK-VTWEG	VBAK-SPART	VBAK-ERDAT	VBAK-ERNAM
VBAK-VBELN	VBAK-AUART	VBAK-VKORG	VBAK-VTWEG	VBAK-SPART	VBAK-ERDAT	VBAK-ERNAM

- Sales Order Number should have a hotspot functionality, clicking on which should display the sales order using the transaction VA03.

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Hands-On Activity 1 (Cont.)

Activity Details: Create an executable program with a selection screen	
Output	**Displaying a List in the ALV Grid Control**: Use the method ‘SET_TABLE_FOR_FIRST_DISPLAY’ of the class ‘CL_GUI_ALV_GRID’ to display the data.

EXPORTING	CHANGING
I_STRUCTURE_NAME	IT_FIELDCATALOG
IT_TOOLBAR_EXCLUDING	IT_OUTTAB

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Hands-On Activity 2

Activity Details: Create an executable program with a selection screen	
Problem Statement	The selection screen should have the following selection criteria. • Select options: – S_MATNR type MARA-MATNR – S_MTART type MARA-MTART • Validate the selection screen parameters, S_MATNR and S_MTART, and display appropriate error messages in case of invalid entry. • Add two Radio buttons RB_LIST and RB_GRID with default to RB_GRID. Selection of radiobutton should show the ALV using SALV class in grid or list mode. Processing Logic: Fetch all the records from table MARA for the first 20 columns that match the corresponding selection criteria.

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Hands-On Activity 2 (Cont.)

Activity Details: Create an executable program with a selection screen	
Output	• Display the data from MARA Table using CL_SALV_TABLE class as per the selection-on-selection screen. • If no records are found for the specified selection criteria, give a proper message and return to selection screen. • For CL_SALV_TABLE – Action/Function which needs to be enabled: – Add Refresh button on the ALV Toolbar - In User command event, perform refresh of data on the ALV when REFRESH button is pressed. – Add hotspot on MARA-MATNR clicked t-code - MM03 – Add option to save layout – Button on Toolbar using SALV methods to navigate to Last Customer who purchased (VBAP-MATNR) this material (XD03) **Note:** Refer SALV *DEMO* programs for reference. Capture specific exception using TRY-ENDTRY.

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Summary

Here are the key learning points of the module.

- Pre-defined methods and events, security, ALV grid control, and reusability are the advantages of creating ALV using CL_GUI_ALV_GRID class.
- The new object model SAP NetWeaver 2004 is consistent and uses a single class for handling tabular data.

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